

Classical Cryptography Techniques

Learning Objectives

- Understand classical ciphers
- Perform encryption/decryption
- Analyze strengths & weaknesses

Quick Revision

- **1. Quick Revision (5–7 mins)**
- What is cryptography?
- Plaintext vs Ciphertext
- Encryption vs Decryption
- Key concept

What are Classical Ciphers?

- Historical encryption methods
- Foundation of modern cryptography
- Mostly symmetric key

Caesar Cipher

◆ Concept

- Each letter is shifted by a fixed number

◆ Example

Plaintext:

HELLO

Shift: +3

Ciphertext:

KHOOR

Caesar Cipher

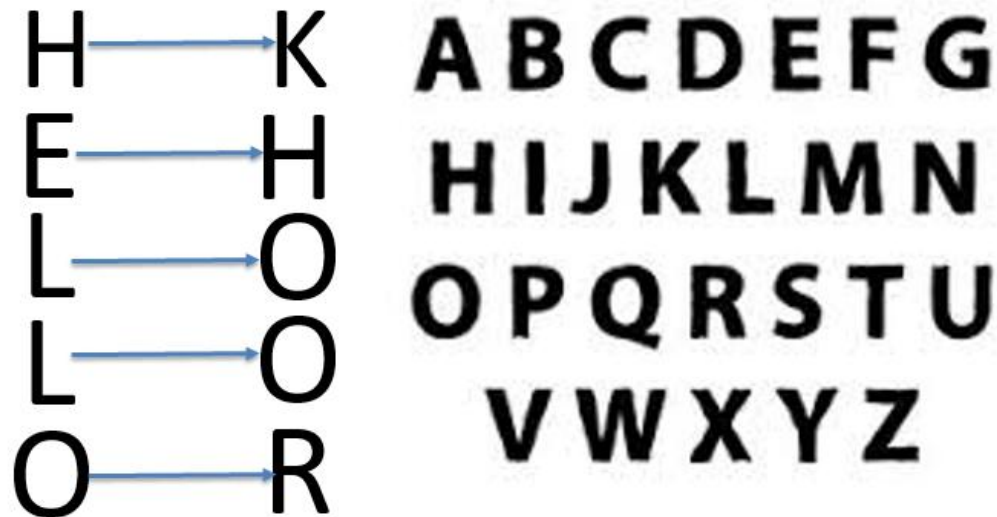
- **Formula (simple form)**
- Encryption: (letter + shift)
- Decryption: (letter – shift)

Instant Trick  **Use this formula in mind:**

- **Encryption:** move forward
- **Decryption:** move backward

Caesar Cipher

- Each letter is shifted by a fixed number
- Example: HELLO → KHOOR (shift 3)



Activity for Students

- Decrypt: FYYFHP
- Key: 5
- Plaintext=?

Caesar Cipher Weakness

- Only 25 possible keys
- Easy brute-force attack

Monoalphabetic Cipher

- Each letter maps to another letter

◆ Concept

- Each letter maps to another letter
- Example:
 - $A \rightarrow Q$
 - $B \rightarrow W$

◆ Key Idea:

- More secure than Caesar
 - But still breakable (frequency analysis)
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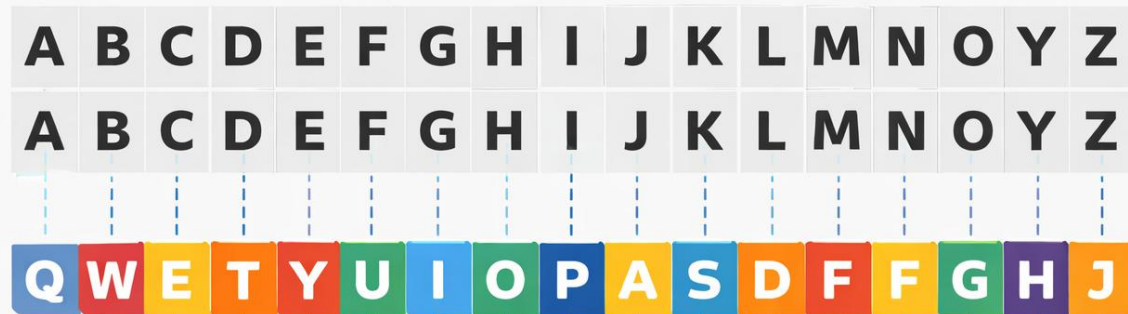
Monoalphabetic Cipher

- What is Monoalphabetic Cipher?
- 👉 It means:
Each letter is replaced by **ONE** fixed different letter
- A will *always* become the same letter
- B will *always* become another same letter
- This mapping does **not change**

Monoalphabetic Cipher

Monoalphabetic Cipher

Plaintext: **HELLO WORLD**



Ciphertext: **T I A A I C K V S**

Plaintext: **H E L L O K V V H S**

Monoalphabetic Cipher



Step 1: Create a Substitution Key

Think of it like a secret alphabet

Normal Alphabet:

B C D E F G H I J K L M N O P Q R S T U V W X Y Z (

Secret (Substitution) Alphabet:

Q W E R T Y U I O P A S D F G H J K L Z X C V B N (

Monoalphabetic Cipher



Step 2: Replace Each Letter

Now take a word:

Plaintext:

HELLO

Convert letter by letter:

Plain	H	E	L	L	O
Cipher	I	T	S	S	G



Final Ciphertext:

ITSSG

Monoalphabetic Cipher

Task for students

Decrypt the given cipher text using monoalphabetic Cipher:

HQLL=Plaintext?

**SECRET ALPHABET**

PLAIN ALPHABET:

A	B	C	D	E	F	G	H	J	J	K	L	M	N
O	P	Q	R	S	T	U	V	W	X	Y	Y	Z	

SECRET ALPHABET:

Q	W	E	R	T	Y	U	I	O	P	A	S	D
F	G	H	J	K	L	Z	X	C	V	B	N	M

Monoalphabetic Cipher

Questions For Students

1. What happens if two same letters appear in plaintext (e.g., “LL”)
2. In monoalphabetic cipher, does one letter map to multiple letters or only one?

Playfair Cipher

- Encrypts pairs of letters
- Uses 5x5 matrix
- More complex than simple substitution

Vigenère Cipher

- Uses a keyword
- Multiple Caesar shifts
- Instead of shifting letters by the same amount every time, the Vigenère cipher uses a **keyword** to change the shift for each letter.

Vigenère Cipher

- **How It Works (Encryption)**
- **Step 1: Choose a Key**
- Example:
Plaintext = **HELLO**
Key = **KEY**
- Repeat the key to match the message length:
KEY → KEYKE

Vigenère Cipher

- **Step 2: Convert letters to numbers**
- (A=0, B=1, ..., Z=25)

Letter	Value
H	7
E	4
L	11
L	11
O	14

Vigenère Cipher

Key:

Letter

Value

K

10

E

4

Y

24

K

10

E

4

Step 3: Apply Formula

Using:

$$C = (P + K) \bmod 26$$

Now calculate:

Plain (P)	Key (K)	Cipher (C)
7	10	17 → R
4	4	8 → I
11	24	9 → J
11	10	21 → V
14	4	18 → S

Ciphertext = RIJVS

Comparison of Ciphers

- Caesar: Very Low security
- Monoalphabetic: Low security
- Vigenère: Medium security

Real-World Connection

- Foundation of AES & RSA
- Used in secure communication

Practice / Assignment

- Encrypt using Caesar
- Break Caesar cipher
- Try Vigenère encryption

Next Lecture

- Cryptanalysis
- Modern cryptography
- Symmetric VS Asymmetric Cryptography
- AES, RSA introduction